

DR. KARLA REYES LEIVA

Ostrava, Czech Republic+420 725 398 646drkarlareyes@gmail.com

ORCID: 0000-0002-3269-0830Scopus ID: 57105041400drkarlareyes.github.io

Google Scholar: 240 citations · h-index: 7 · i10-index: 7 | Scopus: 161 citations · h-index: 6 | July 2026)

PROFESSIONAL SUMMARY

BMD data science researcher, passionate about health tech and applied AI in healthcare. Strong leadership and communication skills. Professional experience in diverse research and engineering task: design and development of applied AI methods for healthcare data, laboratory testing, scientific writing and reporting, teaching, mentoring, and project management. +9 years of research experience across five countries — Brazil, Spain, Germany, the United States, and the Czech Republic. Currently based at the eHealth Lab, VSB Technical University of Ostrava, where I develop contactless radar-based vital sign monitoring systems and I am designing **CycleMotion**, a proposed multimodal longitudinal study of female physiology.

Languages: Spanish (native) · English (advanced) · Portuguese (advanced)

CORE COMPETENCIES & TECHNICAL SKILLS

Signal Processing & Sensing: Radar sensors · Inertial sensors · Time-series analysis · Signal processing · OpenCap/OpenSim · MNE · PyWavelets · SciPy

AI & Machine Learning: Python · TensorFlow · PyTorch · Scikit-learn · Deep learning (CNN, RNN, Transformer) · Computer vision · Neuroimaging · Precision diagnostics · AI for health · OpenCV

Biomedical Engineering: Motion analysis · Medical device standards (IEC 60601, ISO 13485) · Electronics design · Robotics · SolidWorks · EHR data pipelines · Regulatory documentation (SaMD) · Model deployment (Docker, ONNX)

Research & Leadership: Grant writing · Project management · Scientific writing · Mentoring & supervision · Interdisciplinary collaboration · Adaptability · Git · Data visualisation (Matplotlib, Plotly, SQL)

EMPLOYMENT HISTORY

June 2024 – Present

RESEARCHER VSB Technical University of Ostrava , Health Lab, Department of Cybernetics and Biomedical Engineering, Faculty of Electrical Engineering and Computer Science. Ostrava, Czech Republic

- Signal processing techniques for vital sign estimation in freely moving individuals using FMCW radar technology.
- Deep learning methods for human activity recognition and motion characterization using inertial sensors.
- Walking parameter estimation for ankle-foot orthotics using capacitive sensors and edge AI.
- CycleMotion (in development): A multimodal longitudinal study of motion, physiological signals, neurological function, and sleep across the menstrual cycle in healthy women, in collaboration with University Hospital Ostrava.
- Lead author and coordinator of an Erasmus Mundus Design Measures application (2025–2026, decision pending), responsible for full proposal development, consortium design, and international partner coordination.
- Lectures: Applied Artificial Intelligence Methods. Supervision of undergraduate, master, and PhD international students in signal processing and biomedical AI.

Support in project development for the Department of Cybernetics and Biomedical Engineering since February 2026.

March 2025 – Present

LEAD AI DEVELOPER (CONSULTANT) Self Employed. Private US Healthcare Company

- Engineered a clinical decision support system for hormone therapy integrating EHR and longitudinal lab data.
- Designed automated retraining pipelines for continuous model learning and validation.
- Led regulatory documentation toward FDA-aligned Software as a Medical Device (SaMD).

April 2023 – May 2024

ASSISTANT PROFESSOR / RESEARCH FACULTY Universidad Tecnológica Centroamericana (UNITEC). San Pedro Sula, Honduras

- Lectures: Biomechanics and Medical Devices Design (IEC 60601, ISO 1497, ISO 13485), Medical Informatics, Research Project for Biomedical Engineering, Innovation and Technology Management Systems.
- Advised, mentored, and supervised 23+ undergraduate students on biomedical engineering research projects.
- Research lines: Deep learning for redundant robotics, biomechanics for rehabilitation and sports, neuroradiology image processing.

Oct 2022 – March 2023

RESEARCH SCHOLAR University of California Santa Cruz. Computer Vision Lab, Baskin School of Engineering. Santa Cruz, CA, USA

- Deep learning and machine learning for sequential data to predict walking task analysis in visually impaired people using inertial sensor databases.

April 2022 – July 2022

RESEARCH SCHOLAR Karlsruhe Institute of Technology. Computer Vision for HCI Lab & SZS. Karlsruhe, Germany

- Deep learning for object detection to assist visually impaired people in machine interactions.

Oct 2019 – April 2023

PREDOCTORAL RESEARCHER Center for Biomedical Technology. Universidad Politécnica de Madrid, Laboratory of Bioinstrumentation and Nanomedicine. Madrid, Spain

- Developed AI pipelines for multimodal medical signal analysis combining inertial, and image data.
- Designed and validated a motion medical device for visually impaired people (electronic design + validation).
- Published on AI-based motion monitoring, gait analysis, and assistive technology and rehabilitation.
- International collaborations with KIT (Germany) and UC Santa Cruz (USA). Supervised 14+ international students.

Nov 2018 – Sept 2019

ASSOCIATE PROFESSOR Universidad Tecnológica Centroamericana (UNITEC). San Pedro Sula, Honduras

- Lectures: Statistics for Engineering, Introduction to Mechatronic Engineering, Research Project for Biomedical and Mechatronic Engineering, Manufacturing Processes.
- Supervised 20+ undergraduate students. Contributed to university research journal Innovare.

2013 – 2018

ENGINEERING ROLES (PRE-DOCTORAL) Bennet & Bennet / ACQUA SA de CV · Honduras & El Salvador

- Project management engineering and technical sales roles in industrial, clinical, and laboratory equipment sectors.

Jan 2014 – Nov 2015

LABORATORY RESEARCH SCHOLAR Universidade Federal do Paraná. Laboratório de Biomateriais e Eletroquímica. Curitiba, Brazil

- Characterization of metal alloys for biomedical applications: mechanical and electrochemical testing.

AWARDS & DISTINCTIONS

Researcher of the Year — Universidad Tecnológica Centroamericana UNITEC, Honduras 2024

Funded Research Stay — Universidad Politécnica de Madrid, Spain 2022–2023 Awarded to PhD students and professors with exceptional research potential.

Erasmus+ Mobility Grant — European Union 2020–2021

Honduras 20/20 International Scholarship — Honduras 2019–2020 Awarded to high-achieving Honduran students for graduate studies abroad.

Fundación Carolina Doctoral Scholarship — Spain 2019–2023 Awarded to remarkable Latin American academics pursuing a PhD.

OAS–GCUB Master's Scholarship — Organization of American States / Brazil 2014–2016 Awarded to citizens of OAS member states demonstrating academic excellence.

EDUCATION

Doctor of Philosophy (PhD) in Biomedical Engineering May 2024

Universidad Politécnica de Madrid, Spain · International PhD · Scholarship: Fundación Carolina

Thesis: Enhancing gait assessment in assistive technologies for the Visually Impaired integrating inertial sensors.

Supervisor: Prof. Dr. José Javier Serrano Olmedo · Focus: Development of low-cost wearable sensor systems and AI algorithms for gait analysis, rehabilitation monitoring, and mobility assistance in visually impaired people.

Master of Science in Materials Engineering December 2015

Universidade Federal do Paraná, Brazil · Scholarship: OAS & Grupo Coimbra de Universidades Brasileiras

Biomedical Engineer (BSc) June 2013

Universidad Tecnológica Centroamericana, Honduras

GRANT WRITING & FUNDING EXPERIENCE

Erasmus Mundus Design Measures (EMDM) Application. 2025–2026, decision pending

Lead author and coordinator. Responsible for full proposal development, consortium design, and international partner coordination. Submitted to the European Commission.

INFRA Project — Long Cane Biomechanics Platform, Spain. 2023, funded

Co-led a successful infrastructure grant at Universidad Politécnica de Madrid, funded through a Spanish private research infrastructure programme, supporting acquisition of imaging equipment for real-time assistive technology for mobility in visually impaired people.

Assistive Radar-Based Technology for Rehabilitation — Brazil. In submission, 2025

Provided technical support and contributed to writing of a joint research proposal with University of Campinas (UNICAMP), submitted for Brazilian federal health technology innovation funding.

EIT Health Innovation Call — Project CELIA. 2022 & 2023

Contributed to development of two European grant proposals under the EIT Health Innovation Call, supporting the lead coordinator on intelligent assistive technologies for mobility and rehabilitation in elderly individuals.

RESEARCHER PROFILE — SELECTED PUBLICATIONS

Journal Articles (14 published · 1 under review)

- Reyes Leiva KM, et al. Anthropometry and Diagnostic Aware Deep Learning for Exercise Assessment. *Frontiers in Medical Technology*. 2026. DOI: 10.3389/fmedt.2025.1725661
- Reyes KM, et al. SiViS: Simulated Multi-Patient Physiological Clinical States for Advanced Vital Sign Radar Monitoring Research. *Scientific Data (Nature)*. 2026. DOI: 10.1038/s41597-025-06449-9
- Reyes Leiva KM, et al. Estimating Respiratory Rate in Freely Moving Users Using Independent Component and Multi-Resolution Analysis. *Computers in Biology and Medicine*. 2025. DOI: 10.1016/j.compbiomed.2025.110957
- Jaén-Vargas M, Reyes Leiva KM, et al. Effects of Sliding Window Variation in HAR Using Deep Learning. *PeerJ Computer Science*. 2022;8:e1052. DOI: 10.7717/peerj-cs.1052 [82 citations]
- Reyes Leiva KM, et al. Inertial Measurement Unit Sensors in Assistive Technologies for Visually Impaired People: A Review. *Sensors*. 2021;21(14):4767. DOI: 10.3390/s21144767 [32 citations]
- Reyes Leiva KM, et al. Estimation of Spatio-Temporal Parameters of Gait Using Wearable Sensors. *Sensors*. 2023;23(12):5564. DOI: 10.3390/s23125564 [12 citations]
- Namur RS, Reyes KM, Marino CEB. Growth and Electrochemical Stability of Compact Tantalum Oxides. *Materials Research*. 2015;18(S1):91–97. DOI: 10.1590/1516-1439.348714 [35 citations]

+ 7 additional published journal articles · Full list in attached Publications document.

Under Review:

- A Systematic Review for Reporting Bias in Deep Learning-Based Vital Sign Estimation Studies Using Radar. Submitted to *Pattern Recognition*. 2026.

Conference Publications (10+)

- Multi-Objective ICA for Radar Signals Based Heart Rate Estimation. *IEEE EMBC 2025*. DOI: 10.1109/EMBC58623.2025.11253136
- Prediction of Spinal Abnormalities via Deep Transfer Learning. *IEEE ICMLANT 2023*. DOI: 10.1109/ICMLANT59547.2023.10372991
- Finite Element Analysis on Ankle-Foot Orthosis. *IEEE EHB 2024*. DOI: 10.1109/EHB64556.2024.10805597

+ 7 additional conference publications · Full list in attached Publications document.

PROFESSIONAL REFERENCES

Prof. Dr. Marek Penhaker · Supervisor, LERCO, VSB Technical University of Ostrava marek.penhaker@vsb.cz

Prof. Dr. Martin Cerny · Direct supervisor, Health Lab, VSB Technical University of Ostrava martin.cerny@vsb.cz

Prof. Dr. José Javier Serrano Olmedo · Principal Investigator, Lab. of Bioinstrumentation and Nanomedicine, Universidad Politécnica de Madrid josejavier.serrano@ctb.upm.es

Prof. Dr. Claudia Eliana Bruno Marino · Principal Investigator, Laboratório de Biomateriais e Eletroquímica, UFPR, Brazil claudiamarino@ufpr.br

Prof. Dr. Roberto Manduchi · Principal Investigator, Computer Vision Lab, University of California Santa Cruz manduchi@soe.ucsc.edu

Prof. Dr. Karin Mueller · Deputy Director, Center for Digital Accessibility and Assistive Technology, KIT, Germany karin.e.mueller@kit.edu

COMPLEMENTARY EDUCATION & TRAINING

- S.R. AbilityLab Academy — Wearable Sensors in Rehabilitation: Tracking for the Future
- University of Michigan (Coursera) — Python for Everybody specialisation (4 courses) · Intro to AR/VR/MR/XR
- EIT Health — Creative Thinking & Innovation in Healthcare (Trinity College Dublin) · Mobility and Active Ageing (Universidade de Lisboa)
- Universidad Politécnica de Madrid — Autonomous and Connected Vehicle Summer School · Science Communication (UCC UPM)
- ELSEVIER Researcher Academy — Scopus Indexing Criteria · Introduction to Academic Publishing
- COPE — Introduction to Publication Ethics
- WIPO — Patent Application Writing Workshop (San Salvador, El Salvador)